

CAPSTONE PROJECT PITCH

# Next-Gen Music Recommender

on real Yandex listening data (Yambda)

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HSE Master's · NNDL · Module 2

**Real-world operational data (+2 bonus)**

[github.com/ak1232320/nndl\\_capstone\\_2026](https://github.com/ak1232320/nndl_capstone_2026)

## 01 / PROBLEM STATEMENT

# Predict the next track for a streaming user

### What we are solving

Given a 10 M-track catalogue, predict the next track a user will actually want to hear — the central decision a streaming product makes every few minutes.

### Why now

- $\approx 70\%$  of streaming users struggle to discover new music — bad recommendations are the dominant churn driver.
- The streaming market is saturated; subscriber growth has plateaued, so retention is the #1 commercial metric.
- A few % lift in next-track ranking separates catching up from leading.

## 02 / DATA — OUR KEY ADVANTAGE

# Yambda · 4.79 B real listening events from Yandex Music

### Source

- Yambda by Yandex on Hugging Face — released May 2025 (arXiv:2505.22238, RecSys '25).
- 1 M users · 9.39 M tracks · 4.79 B events (**we train on the 50M slice**).

### Why it is strong for this task

- Three signal layers: explicit (likes/dislikes), implicit (plays/skips/replays), and 128-d audio embeddings (82 % of tracks).
- An `is_organic` flag on every event separates user-driven from recommender-driven plays — trains without the recommender-bias loop.
- Real production distributions: heavy-tail catalogue, real cold-start, real session structure.

Real-world operational data → qualifies for the +2 bonus, target = 10 / 10.

### 03 / MODELING APPROACH

## Hybrid: SASRec sequence model + audio-content tower

#### A · SASRec — sequence

- Causal-attention Transformer over the user's listening history (structurally a small GPT).
- Learns transition patterns and predicts the next track from the sequence so far.

#### B · Content tower

- Scores tracks by audio-taste similarity on Yandex 128-d embeddings — complementary to the sequence model.

#### Fusion & why this approach

- Outputs fused with a **validation-tuned weight ( $\beta$ )** — late, not joint, fusion (joint overfit).
- Pure CF loses sequence; pure content ignores behaviour — the hybrid handles both.

# NDCG@10 — ranking quality at the top of the list

### Primary metric

NDCG@10 under the dataset's Global Temporal Split, ranking over the full 629 k-item catalogue (the same protocol as the Yandex paper).

### Results (our harness)

- Best baselines: ItemKNN 0.071 · **SASRec 0.0735** (98 % of the paper, beats all baselines).
- **Hybrid: +9.7 %  $\pm$  2.0 % over SASRec** — robust across 5 held-out user splits.

### Expected business impact

- +9.7 % better next-track ranking → lever on listening time and churn.
- Delivered on free data + one free GPU (near-zero infrastructure cost).

## 05 / SUMMARY & CALL TO ACTION

# What we ship

Real Yandex listening data (Yambda-50M) → hybrid neural recommender: SASRec + audio-content tower, fused with a validation-tuned weight → predict the next track. **SASRec beats every baseline and reproduces the paper to 98 %; late fusion adds a robust +9.7 % ± 2.0 % NDCG@10** across 5 held-out splits.

### What it took / how to reproduce

- Yambda dataset on Hugging Face — public, free.
- One free Kaggle T4 (16 GB) GPU — no paid infrastructure.
- Reproducible: [github.com/ak1232320/nndl\\_capstone\\_2026](https://github.com/ak1232320/nndl_capstone_2026) — one notebook **RUN\_ALL.ipynb** (Run All).